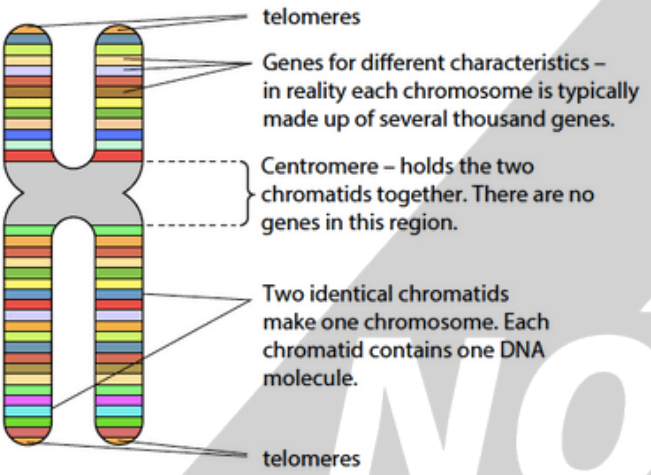
1. The Mitotic Cell Cycle

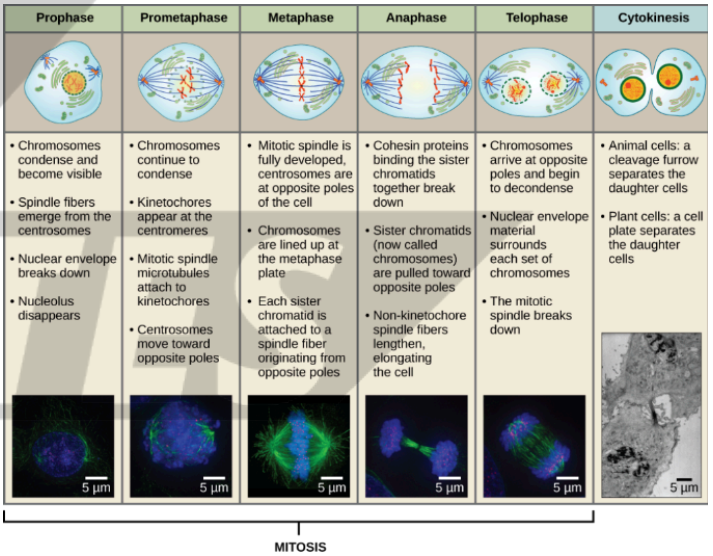
1.1. Structure of Chromosomes

- Chromosome:
  - Threadlike structure containing DNA and genes
- Chromatin:
  - A combination of DNA (acidic) wound around histone proteins (basic)
  - Chromosomes condense, so DNA is tightly packed, making it easier to separate chromatids at the centromere into daughter cells
- Telomeres:
  - Repeated short base sequence at the end of the chromosome (by telomerase)
    - Permit continued replication
    - Prevent the loss of genes
    - Protect ends of chromosomes from being degraded
- Centromere:
  - The region where chromatids are held together



1.2. The Cell Cycle

- 3 phases:
  - Interphase → Nuclear division → Cell division
- Interphase:
  - The cell grows to its normal size after cell division and synthesises important substances, e.g., proteins
  - Three phases: G1 → S phase → G2
- Gap 1 phase (G1):
  - Gap after cell division (cytokinesis) and before S phase
  - 46 chromosomes and chromatids at this phase
  - Producing RNA, proteins, and enzymes prepares for growth and DNA synthesis (S phase)
- S phase:
  - DNA synthesis so each chromosome has two identical chromatids
  - 46 chromosomes and 92 chromatids at this stage
- Gap 2 phase:
  - Gap after S phase and before nuclear division (prepares for mitosis)
  - New DNA is checked, and errors are repaired
  - The sharp increase in tubulin production makes microtubules for the formation of mitotic spindle
- Nuclear division (mitosis):
  - Growth stops temporarily during mitosis
- Importance of mitosis:
  - Replacement of cells
  - Repair of tissue
  - Growth/increase in cell numbers
  - Asexual reproduction / vegetative
  - Maintains / the same number of chromosomes
  - Genetically identical to parents
  - A produces genetically identical daughter cells
  - Cloning of B and T-lymphocytes during immune response
- Phases:
  - Prophase, Metaphase, Anaphase, Telophase



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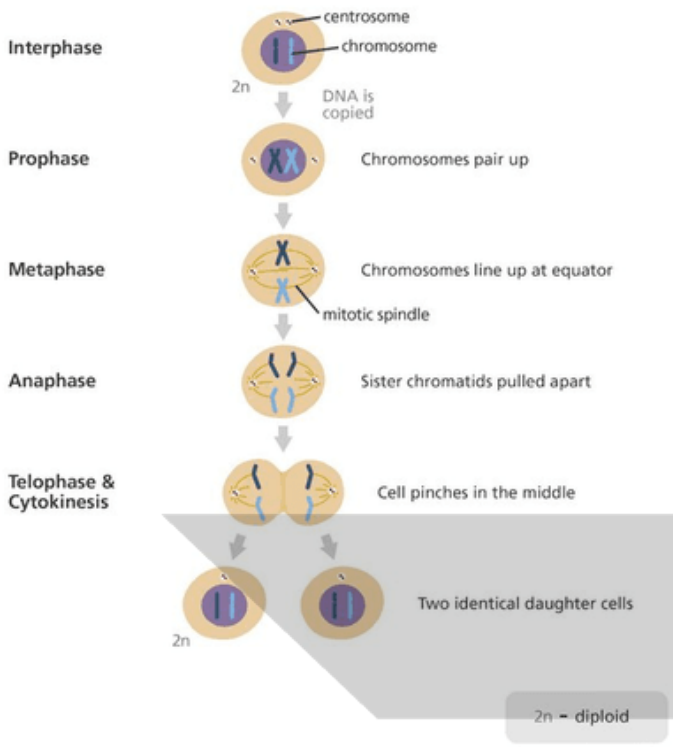


- **Note:** The nuclear envelope breaks down into vesicles during prophase and reassembles when the vesicles fuse to form the envelope back at telophase
- **Cytokinesis:**
  - Division of cytoplasm between daughter cells

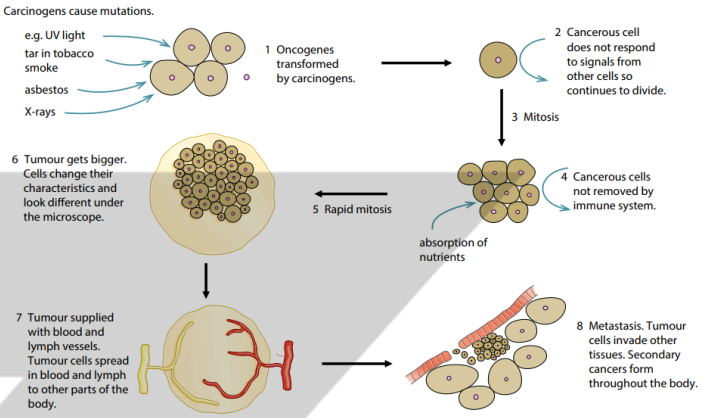
1.3. Stem Cells and Cancer

Stem Cells

- **Stem Cells**
  - Cells that divide repeatedly by mitosis and differentiate into specialized cells or remain as stem cells
  - There are three different kinds:
    - **Totipotent:** Cells that can repeatedly divide to form any other cell in the body, e.g., zygote
    - **Pluripotent:** Embryonic stem cells that lead to the development of the embryo and later the adult; they are not specialized in the placenta
    - **Multipotent:** Adult stem cells that can only produce a few types of cells, e.g., stem cells in bone marrow
  - Their telomeres don't shorten as telomerase is present and active

Cancer

- **Cancer:**
  - Mutation occurs in genes that control cell division, an oncogene that results in uncontrolled mitosis
  - Cancerous cells divide repeatedly and form a tumour, which is an irregular mass of cells
- **Malignant tumour:**
  - Tumours that spread through the body invade other tissues and destroy them
  - These cells break off from the tumour and form secondary growth, known as metastasis.
- **Mutation:** A random change in the base sequence of DNA (gene mutation) or in the structure/number of chromosomes (chromosome mutation)



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